

Stochasticity in evolution

Thomas Lenormand¹, Denis Roze² and François Rousset³

¹ Centre d'Ecologie Fonctionnelle et Evolutive, UMR 5175, 1919 Route de Mende, F-34293 Montpellier cedex 5, France

² Station Biologique de Roscoff, CNRS, Adaptation et Diversité en Milieu Marin, 29682 Roscoff, France

³ Université Montpellier 2, CNRS, Institut des Sciences de l'Evolution, 34095 Montpellier, France

The debate over the role of stochasticity is central in evolutionary biology, often summarised by whether or not evolution is predictable or repeatable. Here we distinguish three types of stochasticity: stochasticity of mutation and variation, of individual life histories and of environmental change. We then explain when stochasticity matters in evolution, distinguishing four broad situations: stochasticity contributes to maladaptation or limits adaptation; it drives evolution on flat fitness landscapes (evolutionary freedom); it might promote jumps from one fitness peak to another (evolutionary revolutions); and it might shape the selection pressures themselves. We show that stochasticity, by directly steering evolution, has become an essential ingredient of evolutionary theory beyond the classical Wright–Fisher or neutralist–selectionist debates.

A long history of debate

The role of stochasticity in evolution has always been a source of debate. For instance, Fisher's view of evolution involved a flux of numerous beneficial mutations of small effect occurring in large populations [1], whereas Wright put forward the importance of stochasticity in small subpopulations in creating combinations of individually deleterious alleles which together have a beneficial effect [2,3]. Similarly, the neutralist–selectionist debate was articulated around the relative importance of genetic drift and selection [4]. In addition, the debate over the 'adaptationist programme' focussed on the role of historical contingency versus necessity [5]. In many cases, debates have concentrated on the role of drift, a particular form of stochasticity.

Simple rules of thumb are often used to determine whether drift is important or not. For example, drift can overwhelm selection in small populations ('is the product of population size N and intensity of selection s greater or less than one?') or migration among subpopulations ('is the product of population size N and migration rate m greater or less than one?'). However, such rules could be misleading in that they fail to distinguish different ways by which stochasticity can affect evolution. Alternatively, a more global approach to the role of chance in evolution has been to focus on the outcome of evolutionary change: if evolutionary change is predictable or repeatable, it would indicate that chance plays only a minor role (Box 1). A limitation of this approach, however, is that comparing outcomes is not sufficient to fully evaluate the importance of stochasticity in evolution. First, the trajectory (and not only the outcome) can be of interest (e.g. it can determine

the genetic basis of adaptation and the rate of adaptation). In fact, the importance of 'history' in evolution has been stressed repeatedly [6,7], based on the idea that because it accumulates over time, evolutionary change is necessarily path dependent and nonrepetitive in all details. A similar situation occurs, for instance, in mathematical optimisation of a unimodal function (which would represent the climbing of an adaptive peak). Several algorithms can be used, some being stochastic (trial and error) and some being deterministic (e.g. the method of steepest ascent). Even if, quite predictably, all algorithms should converge toward the maximum, the path taken and the speed to reach the peak will depend on the algorithm used. In addition, stochasticity can change the position of the fitness peak itself (as though, in our mathematical analogy above, a different function would be maximised depending on the algorithm used); we will develop this idea below.

As we will see, stochasticity has more profound effects than simply making evolutionary trajectories less predictable, or less repeatable. These effects are becoming essential ingredients of evolutionary theory in various domains, from the evolution of life histories to speciation and the evolution of sex. To have an overall perspective on these effects, it is necessary to distinguish the different forms of stochasticity that affect evolution, and the biological processes in which they play an important role. This is the purpose of this review.

The different forms of stochasticity

The notion of chance in evolutionary biology is often quite specific, referring to independence with regard to adaptation [8,9]. Indeed, mutations occur independently of their effect; often, the reproductive success of each individual is,

Glossary

Deterministic: in a model, a process is deterministic when variables take a unique and nonvariable value at each time step.

Deterministic chaos: refers to a deterministic process where very close initial conditions can lead to extremely different outcomes.

Distribution of mutational effects: the statistical distribution giving the probability density that a mutation of a given effect occurs.

Genetic drift: fluctuation of allele frequencies caused by the stochasticity of individual life histories.

Hysteresis: a situation where the outcome of a deterministic process depends on the historical path.

Mutational meltdown: self-reinforcing process in which deleterious mutations fix by drift, which decreases population size, which in turn increases drift and the chance that new deleterious mutations will fix, and so on.

Shifting balance theory: theory of adaptation proposed by Wright involving a shifting balance between evolutionary forces in three phases, with a predominant role of drift, selection and migration.

Stochastic: in a model, a process is stochastic when variables follow a distribution with a nonzero variance.

Corresponding author: Lenormand, T. (thomas.lenormand@cefe.cnrs.fr).

Box 1. Is evolution predictable or repeatable?

To what extent evolution is repeatable is often invoked in thought experiments such as ‘replaying life’s tape’ [7,114]. More precisely, it can be studied with experiments involving replicated evolution in similar environments either in the laboratory (e.g. [115–118]) or nature (e.g. in similar islands [119] or lakes [120]). Clearly, the development of experimental evolution – especially in microbes – in the last decades is an important step toward understanding to what extent evolutionary change can be predicted. However, it is clear that most often, not all forms of stochasticity are included in these experiments (e.g. the effect of environmental stochasticity can be reduced under laboratory conditions).

The question of repeatability or predictability can be asked at the phenotypic level (convergence of an adaptive trait), but also at the genetic level. In particular, the genetic architecture of adaptation might vary depending on the order of appearance of mutations [121,122] or the tinkering of existing genetic material [6,123,124]. For instance, convergent evolution of wing length is achieved by different means in New and Old World populations of *Drosophila subobscura* [125]. However, exactly the same adaptive solution can sometimes occur repeatedly. For instance, the same mutation causes resistance to organophosphorous insecticides in different genera of mosquitoes [126]. Often, a limited number of trajectories might in fact be possible [127,128].

in large part, independent of its genotype; and often, the environment experienced by each individual changes independently of its genotype (we elaborate on these three ideas below). As a consequence, evolution is conveniently described as a stochastic process. Of course, this representation has nothing to do with the claim that the natural world is, *in fine*, deterministic or not. The latter question seems empirically unanswerable anyway, because a deterministic process might easily appear random (e.g. deterministic chaos [10]) and a stochastic process might easily appear deterministic (e.g. statistic regularity with large numbers, as in thermodynamics; Box 2). It is also important to underline that this evolutionary notion of chance is not exactly overlapping (although often consistent) with the more statistical definition of stochasticity based on the idea of sampling (i.e. stochasticity is due to our ignorance of cause or to the coincidence of multiple and independent causes). In models, deterministic and stochastic processes are often simply distinguished by the fact that the latter imply describing some variables, namely the stochastic ones, by a distribution instead of a unique value.

Notwithstanding these different definitions of chance, there are various types of stochasticity that can be considered and that might have different or overlapping effects on evolution. They have especially been categorised

Box 2. Thermodynamics and evolution

The analogy of the role of stochasticity in evolution and thermodynamics has a long history tracing back to Fisher [1] and Schrödinger [129]. For instance, Iwasa [130] proposed an analogy based on tools to analyse the global stability of dynamical systems. Intuitively, one can see that drift spreads populations over a broad distribution of allele frequencies, just as temperature can cause a physical system to stay in a state of high energy. On the contrary, selection makes this distribution concentrate on high fitness (or low energy in the analogy). The two processes equilibrate and, in fact, it is possible to define a ‘free fitness function’ by analogy with free energy in thermodynamics, which is maximised at this equilibrium between selection, mutation and drift.

in reference to factors contributing to population or species extinction (e.g. [11–13]). We group stochastic effects into three broad categories that correspond to different scales of observations: stochasticity of mutation at the gene level, stochasticity of life histories at the individual level and stochasticity of the environment at the scale of populations.

The stochasticity of mutation and variation

By and large, mutations are random: they occur independently of their phenotypic effects, they are not directed and they do not occur more frequently when they are advantageous [14]. Importantly, the fact that these mutations occur at random in this sense does not mean that beneficial and deleterious mutations are equally likely: it is well established that most newly arising mutations are deleterious [15], presumably because of the history of past adaptation. This situation is well summarised in geometrical models of adaptation [16,17], where mutations cause unbiased phenotypic effects, but where the average fitness effect of mutations becomes more and more deleterious the closer a population approaches a phenotypic optimum.

When a mutation is limiting (i.e. when the product of population size N and mutation rate μ is not large), the effect and order of appearance of mutations will be very variable. The beneficial mutation that occurs first will take a population on a particular track toward a particular adaptive ‘solution’ or ‘peak.’ The stochasticity of the occurrence and effect of mutations raises different kinds of questions. First, if there are multiple adaptive peaks, alternative solutions might be reached depending on the order of appearance of mutations. Second, even if there is a single peak, the genetic architecture of adaptation can still vary depending on the sequence of mutations available. However, regarding the stochasticity of mutational effects, the theory lags far behind the data: the stochasticity of mutational effects is rarely incorporated in evolutionary models, which typically consider a very large number of loci each with small and identical effect (the so-called infinitesimal model) or the fate of a mutation with a given and fixed effect (see Ref. [16] for a historical account). Incorporating a distribution of mutational effects (which can include biased mutations [18]) or considering the order of appearance of different mutations is not necessarily an issue in an infinite population which would include all possible mutants (and therefore widely distant phenotypes), but doing so in finite populations is a critical step toward a more quantitative theory of adaptation [19,20], a better understanding of the role played by deleterious mutations (e.g. with Muller’s ratchet [21]) and a more precise description of the genetic architecture of phenotypic traits [18,22].

The stochasticity of life histories

All evolutionary models describe, at least implicitly, the life cycle of individuals. Within this life cycle, several events take place (notably birth, reproduction and death). In general, many other details are considered, such as for sexual individuals mating, syngamy and meiosis. At the usual scale of observation of population biologists, these

life-history events are best described by independent stochastic events for each individual. For instance, the survival of a particular individual from time t to $t + 1$ will be described by a probability. Likewise, the number of offspring produced by a given individual as well as the number of gametes of given genotypes produced by recombination and segregation will be drawn from probability distributions (most commonly a Poisson and a multinomial distribution, respectively, in these two examples). This type of stochasticity can cause random variation in population sizes, which is then referred to as demographic stochasticity. In population genetic models, the effect of this stochasticity of life histories is to introduce a variance in the change in genotype frequencies from one generation to the next, the magnitude of which will depend on the number of individuals (Box 3). This is the usual concept of 'genetic drift.'

However, because the concept of genetic drift encapsulates many sources of stochasticity within a life cycle, it

Box 3. Two standard models of genetic drift

Due to the stochasticity of life histories, the number of offspring produced by an individual is a random variable. Consequently, when an allele is present in a finite number of individuals, its frequency in the next generation is also a random variable. Classical population genetic models are commonly used to investigate the consequences of this source of stochasticity (termed 'genetic drift').

The Wright-Fisher model [1,2]

This standard model represents discrete, nonoverlapping generations. Each generation, individuals produce offspring and die. To form the next generation, a given number of individuals are sampled with replacement from the parents, independently of each other. This corresponds to a multinomial sampling of genotypes. For example, if the population consists of $2N$ haploid individuals of two types (say a and A), and if p is the frequency of A in a given generation, the number of A individuals in the next generation follows a binomial distribution with parameters $2N$ and p . Thus, the frequency of A in the next generation has variance $pq/2N$ (where $q = 1 - p$). Many models have investigated the effects of various deviations from this idealised situation. In several cases, the variance in the change of allele frequency takes the form $pq/2N_e$, where the variance effective size N_e is a function of the parameters of the model (selfing rate, dispersal rates, etc.) but is independent of allele frequencies (e.g. [131,132]). In this case, standard results of the Wright-Fisher model (concerning e.g. fixation probabilities, allele frequency spectra) still apply, replacing N by N_e . Note that in some cases, N_e represents an asymptotic effective size, corresponding to an average over different states in which the population can be found. Indeed, concepts of effective size are useful mostly when there are processes with different timescales, a slow process of drift depending on total population size and fast processes of fluctuation of other demographic properties of the population (e.g. changes in age classes, migration between demes).

Moran's model [133]

This model represents a population in which, at discrete time points $t = 1, 2, 3, \dots$, an individual is chosen to reproduce and another one to die (it is thus one of a range of birth-death models that have been extensively studied in the theory of stochastic processes). Generations are overlapping, and the mean life span of an individual in a population of $2N$ individuals is $2N$ time steps. It can be shown that, if p is the frequency of type A at a given time, the variance of its frequency $2N$ time steps later is pq/N . This suggests that one can define an effective population size for the Moran model as $N_e = N/2$. Indeed, several classical results of the Wright-Fisher model take the same form in Moran's model, replacing N with $N/2$.

comes in many flavours. For instance, it is often thought that there is virtually no drift in an extremely large population. This is not necessarily the case: if the number of offspring of each individual is drawn from a Poisson distribution, a particular allele carried by a given individual can be lost from one generation to the next. Because of this stochasticity in offspring number, the probability of fixation (or more precisely the chance to escape extinction) of a single beneficial mutation of fitness effect s will never exceed a certain threshold, no matter how large the population is. For instance, this threshold is $\sim 2s$ [23] when offspring number is Poisson distributed. Thus, drift still plays a role in large populations.

Furthermore, stochasticity can occur 'locally,' even in an infinite population, for example if the population is subdivided into demes of small sizes [24]. In that case, it is often useful to think in terms of 'local drift' generated by stochasticity within each deme, even if the changes in genotype frequencies in the whole population are deterministic. In this view, inbreeding can also be regarded as a form of local stochasticity. Finally, the fate of an allele could be largely determined by the random occurrence and spread of beneficial mutations at linked loci: a hitchhiking event can suddenly drive such a linked allele to high frequency or cause its loss. Contrary to drift, this process termed 'draft' might be stronger (and even dominate drift) in larger populations where more beneficial alleles can sweep, but its effect will be important mainly when recombination is low [25–27]. Overall, because of population structure and genetic linkage, the stochasticity of individual life histories can be important even in large populations.

The stochasticity of environmental change

The previous section dealt with stochasticity affecting each individual independently. However, environmental changes can also affect all individuals in a population. Environments rarely remain constant, but fluctuate over time in a periodic fashion (seasonal changes) or in a much less predictable manner (weather). Abiotic environmental stochasticity (e.g. frost, fire, volcanic eruptions or asteroid collisions) might come first to mind, but biotic environmental stochasticity is extremely frequent as well. For instance, some demographic regimes can be particularly erratic [10], especially when many species are interacting [28], so that the presence of parasites, predators, prey and even conspecifics can be quite unpredictable for any individual (this idea has also been described as 'biotic drift' [29]). Biotic and abiotic environmental stochasticity can also combine. For instance, by continuously regenerating 'empty' habitats, perturbations play a prominent role in maintaining dynamical demographic equilibria at different scales (metapopulation dynamics [30], succession [31]).

The timescale of environmental change is of particular importance. If environmental fluctuations are very rapid relative to generation time, they are averaged out, resulting in little environmental stochasticity for the organism considered. Only fluctuations arising at longer timescales will matter. Similarly, individuals might be affected by environmental stochasticity only through 'coarse' and not through 'fine'-grained spatial heterogeneity (relative to dispersal distances) [32,33].

Environmental change is not always stochastic in an evolutionary sense. It is important to underline that this is not because some of the causes of environmental variation are well known (e.g. the North Atlantic oscillation, which has a major impact on North Atlantic ecosystems [34]). Rather, like in the case of mutation or drift, environmental change can be considered stochastic when it is not 'directed' and when it occurs independently of adaptation of organisms. However, unlike mutation and drift, environmental change can often be 'directed,' as exemplified by population cycles driven by natural selection [35] or by 'niche construction' processes [36]. As in the case of mutations, it would be desirable to characterise the distribution of fitness effects of environmental changes. Yet, this objective and the methods to reach it remain quite elusive today.

Stochasticity as a cause of evolution

We have distinguished different types of stochasticity at the gene, individual and population levels. Now, we will consider how these different forms of stochasticity influence evolution. However, our emphasis will be on the biological situations where stochasticity matters, rather than on listing the effect of each type of stochasticity separately. We distinguish four broad situations (sketched in Figure 1).

Maladaptation is everywhere

Natural selection has produced impressive adaptations, yet selection is not all-powerful. In particular, all forms of stochasticity impose a limit on adaptation [37,38]. Simply because of the history of past adaptation, a new nonneutral mutation or a new random environmental change is likely to be deleterious and to impose a burden on populations (Figure 1a). These effects illustrate clearly that stochasticity is an important source of maladaptation.

However, the effect of drift is more ambiguous. In an infinite population, the frequency of a deleterious mutation is determined by a balance between the mutation rate (μ) and the intensity of selection (s). This equilibrium frequency is approximately μ/s in a haploid population. It

is often thought that drift will increase the average frequency of these mutations, but this is not always the case. For instance, in a diploid population, drift has little impact on the expected frequency of lethal mutations unless these are very recessive [39]. More strikingly, the frequency of mildly deleterious alleles might even be lowest at intermediate population size provided their dominance coefficient is lower than 1/3, through what is known as 'purging by drift' [40]. Inbreeding caused by local drift can also decrease the frequency of recessive deleterious alleles ('purging by inbreeding').

Similarly, in most cases, drift decreases the chance that a beneficial mutation might fix. However, this is not systematically true. For instance, a recessive beneficial mutation could be more likely to fix when there is some form of local stochasticity such as inbreeding or population structure [41,42]. In fact, the most important maladaptive consequence of drift is that it generates a 'drift load,' either by generating frequency fluctuations around a balanced polymorphism [43] or by allowing deleterious mutations to fix [44]. The latter is revealed by heterosis (hybrid vigour) when crossing individuals from distinct populations (e.g. [45]).

Even when the expected frequency is only weakly affected by drift, the frequency distribution of deleterious mutations is strongly affected by drift. As a consequence, we would expect stochasticity to matter in evolutionary situations where deleterious mutations play a prominent role [46,47]. We can highlight three important cases. First, deleterious mutations can influence the evolution of genetic systems. Because such mutations tend to be recessive, they generate inbreeding depression and they can be masked in diploids. This means that deleterious mutations might promote the evolution of mechanisms of inbreeding avoidance and diploid life cycles, respectively. However, this effect must be balanced by the fact that deleterious mutations can also be purged more efficiently under inbreeding [48] or haploid selection [49]. Because stochasticity affects both the magnitude of inbreeding depression (which is lower in smaller populations) and the efficiency of selection against deleterious alleles (the mutation load increases as population size decreases) [40,50], it should affect the balance between these two forces. However, the effects of stochasticity on the evolution of mating systems and life cycles have been little explored.

Second, when the load they generate becomes large, deleterious mutations can have a demographic impact. Both the extinction of asexual populations by Muller's ratchet [51] and the extinction of small sexual populations by mutational meltdown [52] could occur when drift contributes to the accumulation of deleterious mutations. This demographic decline might particularly occur in marginal populations, and could contribute to the evolution of the niche [53]. However, demographic variations are also very sensitive to other forms of stochasticity. Obviously, extinctions can also be caused by environmental stochasticity [11,12]. Less evidently, introducing stochasticity of the effect of mutations into models (in particular the possibility of compensatory mutations) has been shown to alter meltdown dynamics [54–56].

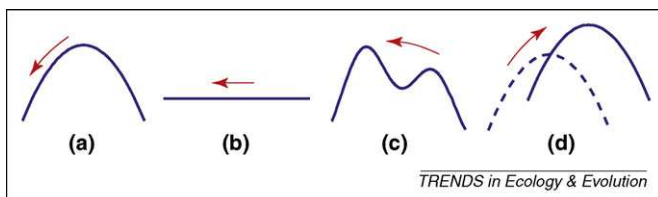


Figure 1. This figure sketches the four broad situations where stochasticity matters in evolution. (a) First, stochasticity can drive evolution down adaptive peaks (*sensu* fitness functions), and this maladaptation can shape important features of organisms. (b) Second, stochasticity freely drives evolution on flat fitness landscapes. Beyond the neutralist–selectionist debate, such ‘evolutionary freedom’ might be important in some cases to understand the extraordinary diversity and sometimes eccentricity of forms and patterns. (c) Third, stochasticity could allow populations to jump from one adaptive peak to another. This idea of ‘evolutionary revolution’ has, however, been a source of continuing controversy. (d) Fourth, stochasticity might itself generate specific selection pressures and orient adaptive evolution (i.e. with stochasticity, the fitness landscape differs as sketched by the dashed lines). This last issue is perhaps less well appreciated, at least experimentally.

Finally, the strongest effect of the accumulation of deleterious mutations is perhaps seen at the genome level, where it might account for the evolution of a diversity of features, especially in higher eukaryotes with low population sizes. First, it certainly contributes to the degeneration of Y chromosomes [46]. Second, the fixation of deleterious mutations might lead to the evolution of unnecessarily complex structures. For instance, the complexity observed in the genome of multicellular eukaryotes relative to the streamlined genomes of prokaryotes (introns, flanking sequences, etc. [57]) as well as in the genetic networks of eukaryotes (redundant regulation [58]) might have originated from the accumulation of slightly deleterious modifications. Of course, an increase in genome size and complexity might have triggered subsequent evolutionary changes and opened new routes of evolution, as exemplified by the evolution of new functions by gene duplication [59].

Evolutionary freedom

Selection imposes strong phenotypic constraints. This is most clearly seen in proteins that are conserved over long timescales and in widely distant organisms. The absence of change in a sequence is most often interpreted as purifying selection, whereas most variation can be interpreted as being neutral and therefore unimportant for adaptation. However, there are different ways by which neutral mutations might play an important role in evolution (besides accounting for patterns of polymorphism at silent sites) and, obviously, the fate of a neutral mutation is entirely stochastic (Figure 1b). The first idea is that neutral mutations are in fact often conditionally neutral (i.e. only neutral in some environments and genetic backgrounds) [60]. This might lead to the free accumulation of cryptic genetic variation, which could become an essential source of adaptive variation when the environment changes [61]. Conditionally, neutral mutations might also result in free accumulation of different neutral alleles in different populations, which could then prove incompatible when combined into the same genetic background: evolution on holey landscapes might in principle cause the evolution of reproductive isolation [62]. Overall, these different views rely on the fact that mutations are not unconditionally neutral, beneficial or deleterious [63].

In many situations, selection imposes a phenotypic constraint, but this constraint is compatible with a large range of equivalent solutions. The usual metaphor is that of a fitness landscape with a ridge. Chance is then the main factor determining which precise phenotype(s) a population will evolve. This is, for instance, the case when the main selection pressure is to ensure matching between two components (for instance, two different RNA sequences can have the same stem-loop secondary structure as long as bases pairing up in stems are complementary). An important case is the evolution of the aesthetics of mate preference: chance might often well explain the capriciousness of female choice and the extravagance of male traits (or vice versa). Sexually selected traits in males have often no clear adaptive value with respect to environmental conditions or male–male competition. Rather, they seem to only evolve to match some arbitrary female preference. Several models have been proposed to explain the

evolution of preference [64], but this idea is mainly captured in ‘runaway’ models where the direction of evolution depends on initial conditions and where drift can easily perturb equilibria [65,66]. Other models have pointed out that the evolution of display traits might nevertheless be restricted to stimuli that can be perceived efficiently by females, whose perception has evolved for many other reasons [67], or to costly traits that honestly signal genetic quality [68]. Obviously, clear signals and cheap preferences are more likely to evolve, but this does not account for the extraordinary variety of male displays. The observation that different ‘themes’ seem more frequent in different groups (e.g. loud vocalisation in frogs, bright colours in birds, etc.) might suggest that preference could somehow be predictable. However, even when there is a theme, there is still room for improvisation: males of different paradise bird species have different displays (shape and colour) around the same theme (long and colourful tails). Clearly the field of possible displays and preferences seems much less constrained toward specific solutions compared with adaptations to environmental conditions. A similar situation occurs with the evolution of the diversity of some aposematic warning patterns, which have to be conspicuous for the predator but can otherwise vary to a great extent in their details [69,70]. Importantly, this relative ‘evolutionary freedom’ could allow sexually selected traits to play a prominent role in speciation [71,72]. These different examples show that the role of evolution on flat landscapes extends well beyond the question of patterns of neutral polymorphism.

Evolutionary revolutions

Sometimes, natural selection can drive a population to a local fitness peak and subsequently prevent it from moving away, even if the peak is lower than other peaks. Because it perturbs allele frequencies, drift has long been recognised as a possible mechanism allowing jumps from one peak to another (which is metaphorically comparable to a genetic ‘revolution’; Figure 1c). This might occur when population size is low, even transiently (e.g. founder effects and bottlenecks). This scenario has been advocated as a mechanism of speciation (founder effect speciation [73,74], ‘punctuated’ speciation [75]) and adaptation (shifting balance theory [2]).

However, these views are controversial. Experimental work has largely failed to demonstrate that bottlenecks cause strong and long-lasting pre- or postmating isolation (e.g. [76]). Similarly, there is little evidence showing that a particular adaptation occurred via the shifting balance process [3]. Nevertheless, the evolution of some traits (e.g. the distribution of chromosomal rearrangements in some mammals [77]) is well explained by a combination of stochastic or historical factors.

It might be unlikely that a genetic revolution is driven by drift alone, but other mechanisms including other forms of stochasticity could cause a peak shift. In fact, it is important to underline that an abrupt switch between simultaneously stable equilibria can occur deterministically, a phenomenon known as hysteresis. Several such theories have been proposed. In the context of speciation, postzygotic isolation might occur via fit intermediates through the accumulation of Dobzhansky–Muller incom-

patibilities [78,79]. Stochasticity also plays an important role in this theory, but it relies more on the stochasticity of occurrence of mutations in different populations than on drift. For adaptation, several mechanisms different from drift have been proposed to allow populations to cross adaptive valleys (this is often termed a ‘deterministic peak shift’ [80–84]). Most often, these theories involve an additional factor which suppresses the valley separating different peaks (e.g. varying selective conditions) or a feedback loop that magnifies the effect of stochastic perturbations. For instance, the evolution of specialisation to different habitats can be particularly sensitive to demographic disturbance, when there is a strong feedback between adaptation and demography [85].

When stochasticity shapes selection

In this last section, we present three important situations where stochasticity might shape the direction of evolution of particular traits (Figure 1d).

First, environmental stochasticity generates uncertainty in future conditions which can generate a selection pressure to minimise the potential detrimental effects of future and unpredictable conditions. Different strategies can evolve to buffer uncertainty and to adapt to a changing world. First, it is possible to phenotypically adjust to present conditions using plastic or inducible responses (e.g. lac-operon regulation in *Escherichia coli*, stress responses in animals, inducible defences in plants [86]). With such a mechanism, it does not matter whether the environment changes stochastically or not because its present state is directly evaluated by the organism based on specific cues. Of course, these cues might not be completely reliable or could be absent. In this case, ‘robust’ phenotypes might nevertheless evolve to minimise the potential detrimental effects of future and unpredictable conditions.

Whether such robustness can evolve has long been controversial [87]. From a genetic point of view, it is known from the debate between Wright and Fisher in the 1930s that the selection pressure to increase the recessivity of deleterious mutations arising at a particular locus is very weak [88]. However, this does not preclude that more global mechanisms of genetic robustness (e.g. chaperone proteins) could evolve and favour genotypes with the ‘flat-test’ fitness function [89,90]. From a more ecological perspective, the question of robustness has mainly been phrased in terms of the evolution of ‘bet-hedging’ or ‘risk-spreading’ life-history strategies (Box 4). Bet hedging has been classified as either conservative or diversified [91]. Conservative bet hedging corresponds to mechanisms buffering environmental uncertainty (homeostasis, margins of safety, resource storage) without involving a phenotypic polymorphism, whereas diversified bet hedging involves strategies which produce different phenotypes (to avoid ‘placing all eggs in the same basket’). A classic example of diversified bet hedging is the production of both dormant and nondormant seeds to escape extinction after bad years [92,93].

Second, we saw that stochasticity often generates maladaptation and imposes a limit on the rate of adaptation. Hence, it can also create a selection pressure to adapt

Box 4. Bet hedging

To illustrate how bet hedging can evolve, we give a simplified model. Let us consider a situation where, each year, individuals use two reproductive strategies with probability a and $1 - a$. The first corresponds to having a fecundity (or growth rate) that varies depending on the quality of the year ($F[1 + \sigma]$ in the good years and $F[1 - \sigma]$ in the bad years, each type of year being equally frequent). The second corresponds to having a constant but lower fecundity every year $F[1 - \sigma]$. Let us consider a mutant with frequency p using a different strategy, $a + da$. One can determine the frequency change of the mutant in good and bad years separately, yielding

$$\begin{aligned}\Delta p_{\text{good}} &= s_{\text{good}} p(1 - p) + O(da)^2 \\ \Delta p_{\text{bad}} &= s_{\text{bad}} p(1 - p) + O(da)^2\end{aligned}\quad [1]$$

where $w_{\text{good}} = 1 + s_{\text{good}}$ and $w_{\text{bad}} = 1 + s_{\text{bad}}$ are the fitnesses at first order in da of the mutant strategy in each type of year, respectively. It is straightforward to show that the arithmetic average of s_{good} and s_{bad} is

$$da \frac{c(1 - c(1 - a)) - a\sigma^2}{(1 - c(1 - a))^2 - a^2\sigma^2} \quad [2]$$

From a biological point of view, the mixed strategy $a^* = c(1 - c)/(\sigma^2 - c^2)$ is convergence stable (solving Equation 2 = 0), meaning that if the environment is sufficiently variable ($\sigma > c$), it increases average fitness to play a mixed strategy (i.e. bet hedging evolves). From a more methodological point of view, it is interesting to note that the same results are obtained if one considers the strategy with the highest geometrical average fecundity over good and bad years. This very simple example shows that (i) fitness (w_{good} and w_{bad}) is not fecundity and (ii) that variance in fitness plays no role at first order even if fitness fluctuates over generations (although the variance in fecundity does) [134].

Because the selection gradient vanishes at the convergence stable point a^* , it could be concluded that all alleles determining the same expected strategy a^* would be selectively equivalent, whether any one individual uses the two reproductive strategies in exactly the proportion a^* and $1 - a^*$ each year or randomly uses only one of them with probability a^* and $1 - a^*$. In finite populations, the fitnesses of such strategies nevertheless differ by terms of order $1/N$ in favour of the former [135]. One can anticipate that local competition within small demes causes stronger selection for homeostasis (the first strategy), as the fitness at the gene level will depend on the variance of a among demes, which will be a function of deme size rather than of total population size. Similar homeostatic effects of spatial structure operate on sex ratio and other traits [136].

faster to a changing world. The idea that species could evolve to adapt faster is a recurrent theme in evolutionary biology (the so-called evolution of evolvability). It is often criticised on the grounds that it seems teleological (‘natural selection cannot adapt a population for future contingencies’ [94]) or that it seems to involve group selection. Indeed, it is very difficult to evolve mutator alleles in sexual populations [95]. Yet, sex and recombination are thought to evolve in a very similar way, that is to increase the efficacy of selection by increasing variance in fitness (this idea traces back to Weismann [96]). The key requirement for stochasticity in this theory was recognised much later [97], and models explicitly showing that the predicted direction of the evolution of sex depends on stochasticity have been developed only recently (this is known as the ‘Hill-Robertson’ effect; Box 5 [98–100]). Because this phenomenon only requires a minimal number of ingredients (drift, at least locally, and directional selection), it remains among the most convincing general explanations for the evolution of sex. Some evidence in favour of this hypothesis is now accumulating [101], in particular regard-

Box 5. Hill-Robertson effect

The Hill-Robertson effect describes the interference between selection at linked loci [25]. It was first quantified by Hill and Robertson [137] and the term proposed by Felsenstein [97]. Suppose that a beneficial mutation appears as a single copy, whereas another beneficial mutation is segregating at a different locus. The second mutation might occur in the good genetic background (positive linkage disequilibrium [LD], i.e. the two beneficial mutations tend to be found together), or in the bad background (negative LD). Although on average this initial LD is zero (as can be shown by a simple calculation), its variance is not zero owing to the fact that the mutation occurs as a single copy. This variance in LD generates a negative expected LD in the following generations. Indeed, situations where $LD > 0$ tend to be transient (the genotype carrying the two beneficial alleles quickly reaches fixation, after which $LD = 0$), whereas situations where $LD < 0$ tend to last longer (because the two beneficial alleles compete against each other). The same Hill-Robertson effect occurs when the variance in LD is generated by local drift in an infinite population [138], and also works between deleterious alleles recurrently produced by mutation at different loci. As a consequence, sex is favoured because it restores genotypes with extreme fitness (e.g. free of deleterious mutations or combining different beneficial mutations).

ing clonal interference, an extreme form of the Hill-Robertson effect [102] which sets an upper limit on the rate of adaptation in asexuals [103].

Finally, in arbitrarily large populations, individuals might still interact with a finite number of other individuals. For instance, preferential interactions with family members can occur before dispersal, and preferential interactions among neighbours can occur if dispersal is limited. Other examples include interacting cells in a multicellular organism or interacting individuals in insect colonies. The evolution of traits affecting the fitness of individuals beyond their bearer ('social traits,' as classified by Hamilton in four categories: cooperation, altruism, selfishness and spite [104]) depends on the relative distribution of genotypes within and among groups. Hence, local drift can shape the selection pressures that govern the evolution of sociality and life in groups. In particular, differences among groups might result from stochasticity owing to the sampling of a finite number of founders of these groups. Indeed, the direction of selection on social traits depends on genetic associations (relatedness) among these interacting individuals, which can be nonzero even in an infinite population provided it is structured [105]. Relatedness quantifies, in these cases, the effect of local drift, that is of fluctuations in allele frequency among groups.

Genetic differences among these groups of interacting individuals could also result from other mechanisms, such as kin recognition. However, recent models indicate that local stochasticity is also a vital ingredient for the joint evolution of kin recognition and cooperation [106]. Here again, local stochasticity builds the necessary genetic associations (in this case between loci determining the social behaviour and loci involved in kin recognition). The importance of kin selection is sometimes minimised on the grounds that not all populations exhibit strong spatial structure. However, the pertinent scale to measure structure depends on the precise structure of interaction. It might also be minimised on the grounds that few organisms or behaviours are commonly described as being

'social.' However, this view neglects the fact that the definition of a social trait is broad [107] and encompasses traits that are shared by all living organisms (for instance dispersal [108]) or extremely widespread (such as multicellularity [109] or sex ratio [110]). In the latter case indeed, kin selection predicts female-biased sex ratio in groups of related individuals, which matches observations with great precision [111].

Conclusion

The role played by stochasticity has shaped most theoretical debates in evolutionary biology since the 1930 s. There are several important examples where stochasticity plays a major role in evolution. In addition, neutral models are often used to generate null hypotheses against which to test alternatives. However, as we have been discussing, neutrality is only one situation where stochasticity matters. Beyond this debate, it is still difficult to evaluate how the different forms of stochasticity have influenced the evolution of particular traits (emblematic examples being sex [112], dispersal [113]). In fact, the different forms of stochasticity that we have discussed are not equally investigated in evolutionary models. The stochasticity of individual life histories, which causes the usual process of drift, is the most commonly considered. The common view is that drift limits adaptation. This is typically true, and it does so even in very large populations. However, this is only a partial account, as drift also plays a more creative role in evolution. It certainly influences the evolution of a large suite of traits that are often as fascinating (ornaments, social traits and sex, to cite a few) as are delicately designed adaptations. Environmental stochasticity is less often considered. Many models consider environmental change but do not necessarily include the fact that this change is stochastic. As a consequence, how much populations adapt to uncertainty remains unclear. Finally, mutational stochasticity is rarely considered. This situation stems perhaps from the fact that transient dynamics are often neglected compared to 'equilibrium' situations, and that it is perhaps thought that little can be said about this form of stochasticity. Overall, combining the different forms of stochasticity in evolutionary theory remains largely ahead of us.

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